

Objectives:

- Understand and apply the core concepts of the Scrum framework (Roles, Events, Artifacts).
- Fulfill the responsibilities of different Scrum roles (Product Owner, Scrum Master, Development Team).
- Create and manage a Product Backlog using User Stories.
- Plan and execute a development sprint using a project management tool like Jira or Asana.
- Collaborate effectively as a team, using Git for version control and Slack for communication.

Introduction

In this assignment, your team will experience a full software development lifecycle in miniature, using the Scrum framework to guide your work. You will build a small but complete application from the ground up, managing your progress with professional tools like Jira, Git, and Slack.

Project vision: The "Quick-Poll" Application

The goal is to create a simple web application that allows a user to quickly create a poll, share it with their team, and see the results in real-time. It's a tool designed to make team decisions (like "Where should we order lunch?" or "Which feature should we work on next?") fast and easy.

Part 1: Pre-Lab work [MARKED as part of the CC]

This part SHALL be completed before the lab session.

A – Account creation and team setup

- Each team member must create a free account for the following services:
 - Jira or Asana (one project per team).
 - GitHub (each member needs an account).
 - Slack (one workspace per team).
- Within your team, decide who will take on the following roles for this project.
 - Product Owner (PO): Responsible for defining features and prioritizing the backlog. This person is the voice of the customer.
 - Scrum Master (SM): Responsible for ensuring the team follows Scrum practices and for removing any impediments or blockers. This person is the team's facilitator.

- Development Team: The remaining members responsible for building the actual software.

Note: Everyone will still be writing code!

B – Sprint Setup (30 Minutes)

Goal: To set up your tools and define the project scope.

Project Setup:

- GitHub: One person creates a new, empty repository on GitHub and invites all team members as collaborators.
- Jira/Asana: The Product Owner creates a new project using a "Scrum" or "Kanban" template. Invite all team members to the project.
- Slack: The Scrum Master creates a new Slack workspace and invites the team. Create a main channel (e.g., #development).

Product Backlog Creation:

- As a team, brainstorm all the possible features for the "Quick-Poll" application.
- The Product Owner leads the discussion and is responsible for writing these features as User Stories in Jira. The PO will create these items in the "Backlog" section of Jira.
- Your team must create at least 8-10 user stories. (*See the last section for examples*).
- Once the stories are in the backlog, the Product Owner must prioritize them by dragging the most important ones to the top.

Part 2: In-class Work

Sprint 1: Building the first Product Increment

Sprint Goal: *By the end of this Sprint, a user will be able to create a new poll with at least two options, and another user will be able to view the poll and vote for an option.*

Part A: Sprint Planning (20 minutes)

1. The **Scrum Master** facilitates this meeting.
2. The **Product Owner** presents the highest priority user stories from the Product Backlog.
3. The **Development Team** discusses the stories, asks clarifying questions, and decides which stories they can realistically complete within the time allotted for the sprint.
4. Move the selected user stories from the "Backlog" into your first "Sprint" in Jira.
5. Now, for each user story in the sprint, break it down into smaller, **technical tasks**. For example, the story "As a user, I can create a new poll" can be broken into:
 - Task: Design the UI for the 'Create Poll' window.
 - Task: Implement the backend logic to save the poll question and options.
 - Task: Connect the UI 'Save' button to the backend logic.
6. Assign these technical tasks to team members in Jira.

Part B: Development - The Sprint (60 minutes)

1. **Git Branching:** Before writing any code, each developer must create a new branch from main for their specific task. **Do not commit directly to the main branch.**
2. **Work on Tasks:** Develop your assigned features. Update the status of your tasks in Jira by moving them from "To Do" to "In Progress".
3. **Communication:** Use the Slack channel for all project-related communication. If you're stuck, ask for help there!
4. **Daily Scrum (Simulated Checkpoint):** Halfway through the development time, the **Scrum Master** will call for a brief (5-minute) "stand-up meeting". Each team member will answer three questions:
 - *What did I accomplish since the last meeting?*
 - *What will I work on next?*
 - *Are there any blockers in my way?*
5. **Code Integration:** When a task is complete, push your branch to GitHub and create a **Pull Request (PR)**. Another team member must review the code and then merge it into the main branch.
6. **Update Jira:** Once a task is fully complete and merged, move it to the "Done" column in Jira.

Part C: Sprint Review (15 minutes)

1. **Demonstration:** The team demonstrates the working software they built during the sprint to the lab supervisor. You should only show features that are **100% complete**.
2. **Feedback:** The Product Owner gathers feedback from the demonstration to help plan the next sprint.

Part D: Sprint Retrospective (10 minutes)

1. The **Scrum Master** leads this internal team meeting.
2. The goal is to reflect on the process, not the product.
3. Each team member should answer:
 - *What went well during this sprint?*
 - *What could we improve for the next sprint?*
 - *What action items can we create to make that improvement?*
4. The Scrum Master should note down the action items.

Example: User Stories for "Quick-Poll"

Use these as inspiration for your own Product Backlog.

- As a poll creator, I want to enter a question for my poll so that others know what they are voting on.
- As a poll creator, I want to add at least two answer options so that users have choices.
- As a poll creator, I want to click a "Create Poll" button so that my poll is saved and becomes active.
- As a voter, I want to see the poll question and the list of options so that I can make a decision.
- As a voter, I want to select one option and click a "Vote" button so that my choice is recorded.
- As a voter, I want to see the current results of the poll after I vote so that I can see what others have chosen.
- As a poll creator, I want to be given a unique code or link for my poll so that I can share it with my team.
- As a user, I want the application to prevent me from voting more than once on the same poll.
- As a poll creator, I want to be able to see a list of all the polls I have created.